

Hossein Vahidi

Doctoral Researcher · Aalto University
Department of Computer Science

Apr 2022–present

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Research interests

I am broadly interested in the theoretical foundations of distributed and parallel computing, computational complexity, and algorithm design, particularly for graph problems. Recently, I have been studying primitives such as matrix multiplication in parallel computing models, including MPC, Congested Clique, NCC, and low-bandwidth models.

Education

Ph.D., Computer Science — Aalto University Jun 2022–present
Expected 2027; Supervisor: Prof. Jukka Suomela.

M.Sc., Computer Science — Saarland University Oct 2018–Oct 2021
Thesis: Approximate Minimum Directed Spanning Tree in Congest and Congested Clique Models.
Supervisor: Prof. Christoph Lenzen.

B.Sc., Computer Engineering — Shahid Bahonar University 2014–2018
GPA: 18.67/20; Ranked 1st.
Thesis: FIM, a Flow based framework for solving Influence Maximization Problem.
Supervisor: Dr. Hossein Ghasemalizadeh.

Work experience

Doctoral Researcher — Aalto University Apr 2022–present
Distributed Algorithms Group · Department of Computer Science

IT Administrator — Max Planck Institute for Informatics Aug 2020–Mar 2022
Department 1: Algorithms and Complexity

Research Intern — Aalto University Aug 2018–Sep 2018
Distributed Algorithms Group
Distributed recoloring with Jukka Suomela.

Research Intern — Max Planck Institute for Informatics Jul 2018
Theory of Distributed and Embedded Systems Group · D1
Anti-Ramsey coloring with Christoph Lenzen and Saeed Amiri.

Publications

Authors are listed alphabetically by surname, following the convention in the theoretical computer science community.

- [1] **Rectangular Matrix Multiplication in the Low-Bandwidth Model**
Chetan Gupta, Jukka Suomela, and Hossein Vahidi
Preprint. 2026. [arXiv:2606.04652](#). [[arXiv](#)]
- [2] **Complexity of computing the anti-Ramsey numbers for paths**
Saeed Akhoondian Amiri, Alexandru Popa, Mohammad Roghani, Golnoosh Shahkarami, Reza Soltani, and Hossein Vahidi
Journal article. Theoretical Computer Science 1044, 2025, article 115271. [[arXiv](#)] [[Journal version](#)] [[Conference version](#)]
- [3] **Low-Bandwidth Matrix Multiplication: Faster Algorithms and More General Forms of Sparsity**
Chetan Gupta, Janne H. Korhonen, Jan Studený, Jukka Suomela, and Hossein Vahidi
Conference paper. SIROCCO 2025, pp. 333–349. [[arXiv](#)] [[Conference version](#)]

- [4] **Brief Announcement: Low-Bandwidth Matrix Multiplication: Faster Algorithms and More General Forms of Sparsity**
Chetan Gupta, Janne H. Korhonen, Jan Studený, Jukka Suomela, and **Hossein Vahidi**
Brief announcement. SPAA 2024, pp. 305–307. [arXiv \[full paper\]](#) [\[Conference version\]](#)
- [5] **Fast Dynamic Programming in Trees in the MPC Model**
Chetan Gupta, Rustam Latypov, Yannic Maus, Shreyas Pai, Simo Särkkä, Jan Studený, Jukka Suomela, Jara Uitto, and **Hossein Vahidi**
Conference paper. SPAA 2023, pp. 443–453. [\[arXiv\]](#) [\[Conference version\]](#)
- [6] **Approximate Minimum Directed Spanning Trees Under Congestion**
Christoph Lenzen and **Hossein Vahidi**
Conference paper. SIROCCO 2021, pp. 352–369. [\[Conference version\]](#)
- [7] **Complexity of Computing the Anti-Ramsey Numbers for Paths**
Saeed Akhoondian Amiri, Alexandru Popa, Mohammad Roghani, Golnoosh Shahkarami, Reza Soltani, and **Hossein Vahidi**
Conference paper. MFCS 2020, pp. 6:1–6:14. [\[arXiv\]](#) [\[Conference version\]](#) [\[Journal version\]](#)

Academic service

Reviewer: DISC (2023, 2026); OPODIS (2026); PODC (2023); SIROCCO (2021).
Local organizing team member: ALGO 2018.

Teaching

Teaching assistant, Aalto University

Competitive Programming	2024
Programming Parallel Computers	2022, 2023
Distributed Algorithms	2023
Principles of Algorithmic Techniques	2022

Thesis advising

Master’s thesis advisor — Tianxing Wu, Aalto University 2025
Towards the Evaluation of Congested Clique Compatibility on Modern Computers
Supervisor: Prof. Jukka Suomela.

Awards and distinctions

Scholarship, Saarbrücken Graduate School of Computer Science.	2018–2020
Afzalipour Award for Outstanding B.Sc. Student, Shahid Bahonar University of Kerman (university-wide).	May 2018
Iranian national M.Sc. entrance examinations: 4th in Computer Science; 16th in Software Engineering.	2018
15th in the national final, 23rd Iranian National University Students’ Olympiad in Computer Engineering.	Aug 2017
6th place (team), ACM-ICPC Asia Tehran Regional Contest.	Dec 2015

Skills

Programming. C, C++, Python — algorithm implementation.

Parallel computing. OpenMP and CUDA; SIMD vectorization; CPU memory/cache and GPU memory optimization; benchmarking and performance analysis; MPI programming.

Formalization. Lean 4 (working knowledge); formalized distributed and parallel computing results.

Tools. Git; LaTeX.

Languages

English (fluent); Persian (native).